SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	B TECH (AI&DS)	**Date:**	

Code of Conduct

I Vivetha B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: VIVETHA B

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

Scenario

A large-scale search engine receives a user query containing different word forms such as **“analyzing”, “analysis”, and “analytical”**. These words are related to the same concept, but they have different spellings and grammatical forms.

If the search engine stores these words separately, a search for **“analyze”** may not retrieve documents containing **“analysis”** or **“analytical”**. Therefore, the search engine needs a **morphological processing module** that can identify the root and affixes, classify the word formation, and normalize related forms into a common representation.

Step 1: User Query

Suppose a user searches for information about **analyze**.

The search engine may encounter the following word variants in its documents:

analyzing

analysis

analytical

The morphological module processes each word.

Step 2: Process analyzing

The word **analyzing** is the continuous form of **analyze**.

analyzing

↓

analyze + ing

↓

Root + Inflectional Suffix

↓

analyze

- Root/Base: **analyze**
- Affix: **-ing**
- Type: **Inflectional**
- Normalized form: **analyze**

Step 3: Process analysis

The word **analysis** is a noun related to the verb **analyze**. It is not a simple regular inflection like *analyze* + *ed*. The system therefore uses a morphological dictionary/lexical mapping to identify its relationship with **analyze**.

analysis

↓

Related form of analyze

↓

Derivational relationship

↓

analyze

- Root/Base concept: **analyze**
- Type: **Derivational**
- Normalized form: **analyze**

Step 4: Process analytical

The word **analytical** is an adjective related to **analysis/analyze**.

analytical

↓

analysis + ical

↓

Base + Derivational Suffix

↓

analyze

- Root/Base concept: **analyze**
- Suffix: **-ical**
- Type: **Derivational**
- Normalized form: **analyze**

Step 5: Morphological Comparison

Input Word	**Root/Base**	**Affix / Form**	**Type**	**Normalized Form**
analyzing	analyze	-ing	Inflectional	**analyze**
analysis	analyze	lexical/derivational relation	Derivational	**analyze**
analytical	analyze	-ical	Derivational	**analyze**

Step 6: Scenario Result

Suppose the search engine has the following documents:

- **Document 1:** “The researcher is **analyzing** the data.”
- **Document 2:** “The **analysis** gives accurate results.”
- **Document 3:** “An **analytical** method is used.”

The morphological processor converts them into:

analyzing → analyze

analysis → analyze

analytical → analyze

The indexing system therefore stores the common normalized representation:

ANALYZE

When the user searches for **“analyze”**, the search engine can match documents containing **analyzing, analysis, and analytical** because all three are linked to the same normalized representation.

Final Result

Root/Unified Representation = analyze

- **analyzing** → analyze + ing → Inflectional → **analyze**
- **analysis** → related derivational form → Derivational → **analyze**
- **analytical** → analysis + ical → Derivational → **analyze**

Thus, the morphological processing approach improves **retrieval accuracy** by grouping semantically related word forms under a unified representation.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

Scenario

An AI-based sentiment analysis platform receives product reviews containing words such as

“disagree”, “agreement”, and “agreeable”. These words are related to the base concept **agree**, but prefixes and suffixes change their meanings and grammatical forms.

For example, **disagree** expresses an opposite or negative meaning, while **agreement** and **agreeable** generally express a positive or accepting meaning. Therefore, the sentiment analysis system must identify the **prefix, suffix, and base form** and understand how derivational changes affect the meaning.

The morphological parser processes each word and produces a normalized representation for sentiment analysis.

Step 1: Input from Product Reviews

Suppose the platform receives the following review sentences:

Review 1: "I disagree with the quality of the product."

Review 2: "The agreement between the customer and company was good."

Review 3: "The product has an agreeable design."

The morphological module extracts the important words:

disagree

agreement

agreeable

Step 2: Process disagree

The word **disagree** contains the prefix **dis-** and the base word **agree**.

disagree

↓

dis + agree

↓

Prefix + Base

↓

dis- + agree

- **Prefix:** dis-
- **Base/Root:** agree
- **Suffix:** None
- **Transformation:** Derivational
- **Meaning:** Opposite of agree
- **Sentiment effect:** Negative

The prefix **dis-** changes the meaning from **agree** to **not agree/opposition**.

Normalized root: agree

Step 3: Process agreement

The word **agreement** is formed from the base **agree** with the derivational suffix **-ment**.

agreement

↓

agree + ment

↓

Base + Derivational Suffix

↓

agree + -ment

- **Prefix:** None
- **Base/Root:** agree
- **Suffix:** -ment
- **Transformation:** Derivational
- **Meaning:** The state or result of agreeing
- **Sentiment effect:** Generally Positive/Neutral

The suffix **-ment** changes the verb **agree** into the noun **agreement**.

Normalized root: agree

Step 4: Process agreeable

The word **agreeable** is formed using the suffix **-able**.

agreeable

↓

agree + able

↓

Base + Derivational Suffix

↓

agree + -able

- **Prefix:** None
- **Base/Root:** agree
- **Suffix:** -able
- **Transformation:** Derivational
- **Meaning:** Pleasant, acceptable, or easy to agree with
- **Sentiment effect:** Positive

The suffix **-able** changes the meaning and grammatical form of the base word.

Normalized root: agree

Step 5: Morphological Classification"

Input Word	Prefix	Base/Root	Suffix	Transformation	Sentiment
disagree	dis-	agree	—	Derivational	Negative
agreement	—	agree	-ment	Derivational	Positive/Neutral
agreeable	—	agree	-able	Derivational	Positive

Step 6: Semantic Interpretation

The morphological parser identifies **agree** as the common base form.

However, simply converting every word to **agree** is not enough for sentiment analysis because the affixes can change the meaning.

```
            AGREE
           /    \
        dis-     suffixes
         ↓      /      \
     disagree  -ment     -able
                ↓         ↓
            agreement   agreeable
```

The system therefore stores both the **root** and the **morphological information**.

For example:

disagree → agree + dis- → Negative

agreement → agree + -ment → Positive/Neutral

agreeable → agree + -able → Positive

This allows the AI system to understand that **disagree** should not be interpreted as having the same sentiment as **agreeable**, even though they share the same base root.

Step 7: Scenario-Based Sentiment Processing

If a customer writes:

"I disagree with the product quality."

The system detects:

disagree → dis- + agree → Negative sentiment

If another customer writes:

"The agreement with the company was satisfactory."

The system detects:

agreement → agree + -ment → Positive/Neutral sentiment

If another customer writes:

“The product design is agreeable.”

The system detects:

agreeable → agree + -able → Positive sentiment

Thus, morphological information helps the sentiment analysis system interpret the words according to their actual meanings instead of treating all forms as identical.

Final Result

The morphological parsing results are:

1. disagree → dis- + agree → Derivational → Root = agree → Negative

2. agreement → agree + -ment → Derivational → Root = agree → Positive/Neutral

3. agreeable → agree + -able → Derivational → Root = agree → Positive

All three words have the common root **agree**, but their **prefixes and suffixes modify their meanings**. Therefore, the AI sentiment analysis platform should retain both the **root form and affix information** to achieve accurate semantic and sentiment interpretation.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

Scenario

A text analytics pipeline in a **news aggregation system** processes a large number of news articles every day. These articles may contain different forms of the same concept, such as **“govern”, “government”, and “governance”**.

If these words are treated as completely different terms, the topic modeling and clustering system may separate articles that actually discuss the same topic. Therefore, a **morphology-based normalization mechanism** is required to identify the common root, separate the affixes, identify derivational changes, and map related words to a common representation.

Step 1: Input Word

The news aggregation system receives the following terms from different articles:

govern

government

governance

The morphology module processes each word separately.

Step 2: Process govern

The word **govern** is the basic verb form.

It does not contain an additional derivational prefix or suffix.

govern

↓

govern

↓

Base Root

- **Root/Base:** govern
- **Prefix:** None
- **Suffix:** None
- **Derivational layer:** 0
- **Normalized form:** govern

Step 3: Process government

The word **government** is derived from the verb **govern** using the suffix **-ment**.

government

↓

govern + ment

↓

Root + Derivational Suffix

↓

govern

- **Root:** govern
- **Suffix:** -ment
- **Prefix:** None
- **Derivational layer:** 1
- **Transformation:** Derivational
- **Normalized form:** govern

The suffix **-ment** forms a noun representing the institution, system, or result associated with governing.

Step 4: Process governance

The word **governance** is related to the base verb **govern** and contains the derivational suffix **-ance**.

governance

↓

govern + ance

↓

Root + Derivational Suffix

↓

govern

- **Root:** govern
- **Suffix:** -ance
- **Prefix:** None
- **Derivational layer:** 1
- **Transformation:** Derivational
- **Normalized form:** govern

The suffix **-ance** creates a noun representing the process, manner, or state associated with governing.

Step 5: Morphological Breakdown

Input Word	Root	Prefix	Suffix	Derivational Layer	Type	Normalized Form
govern	govern	—	—	0	Base form	**govern**
government	govern	—	-ment	1	Derivational	**govern**
governance	govern	—	-ance	1	Derivational	**govern**

Step 6: Normalization Process

The morphology-based normalization mechanism converts all related forms to the common base:

govern → govern

government → govern

governance → govern

Therefore:

Common Normalized Representation = govern

Step 7: Application to News Topic Modeling

Suppose the news system contains the following articles:

- **Article 1:** "The minister discussed how the country should **govern** effectively."
- **Article 2:** "The **government** announced a new policy."
- **Article 3:** "Good **governance** is important for public administration."

Before normalization, the system may treat the three terms as separate words:

govern ≠ government ≠ governance

After morphological normalization:

govern → govern

government → govern

governance → govern

The topic modeling system can therefore group these articles under a common topic related to **governing/public administration**.

Final Result

The morphology-based normalization mechanism identifies **govern** as the common root for all three terms.

- **govern** → **govern** → Base form → **govern**
- **government** → **govern + -ment** → Derivational → **govern**
- **governance** → **govern + -ance** → Derivational → **govern**

Thus, the final normalized representation for all three input words is:

govern

This normalization helps the news aggregation system group semantically related articles more effectively during **topic modeling and clustering**.

4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation. Derive the morphological structure and normalized root for each input word.

Scenario

A text analytics pipeline in a **news aggregation system** processes a large number of news articles every day. These articles may contain different forms of the same concept, such as **“govern”, “government”, and “governance”**.
If these words are treated as completely different terms, the topic modeling and clustering system may separate articles that actually discuss the same topic.
Therefore, a **morphology-based normalization mechanism** is required to identify the common root, separate the affixes, identify derivational changes, and map related words to a common representation.

Step 1: Input Words

The news aggregation system receives the following terms from different articles:

govern

government

governance

The morphology module processes each word separately.

Step 2: Process govern

The word **govern** is the basic verb form.

It does not contain an additional derivational prefix or suffix.

govern
↓
govern
↓
Base Root

- **Root/Base:** govern
- **Prefix:** None
- **Suffix:** None
- **Derivational layer:** 0
- **Normalized form:** govern

Step 3: Process government

The word **government** is derived from the verb **govern** using the suffix **-ment**.

government
↓
govern + ment
↓
Root + Derivational Suffix
↓
govern

- **Root:** govern
- **Suffix:** -ment
- **Prefix:** None
- **Derivational layer:** 1
- **Transformation:** Derivational
- **Normalized form:** govern

The suffix **-ment** forms a noun representing the institution, system, or result associated with governing.

Step 4: Process governance

The word **governance** is related to the base verb **govern** and contains the derivational suffix **-ance**.

governance

↓

govern + ance

↓

Root + Derivational Suffix

↓

govern

- **Root:** govern
- **Suffix:** -ance
- **Prefix:** None
- **Derivational layer:** 1
- **Transformation:** Derivational
- **Normalized form:** govern

The suffix **-ance** creates a noun representing the process, manner, or state associated with governing.

Step 5: Morphological Breakdown

Word	**Root**	**Affix**	**Type**	**Stem**
activate	act	-ivate	Derivational	act
activation	act	-ion	Derivational	act
reactivation	act	re-, -ion	Derivational	act

Step 6: Normalization Process

The morphology-based normalization mechanism converts all related forms to the common base:

govern → govern

government → govern

governance → govern

Therefore:

Common Normalized Representation = govern

Step 7: Application to News Topic Modeling

Suppose the news system contains the following articles:

- **Article 1:** “The minister discussed how the country should **govern** effectively.”
- **Article 2:** “The **government** announced a new policy.”
- **Article 3:** “Good **governance** is important for public administration.”

Before normalization, the system may treat the three terms as separate words:

govern ≠ government ≠ governance

After morphological normalization:

govern → govern

government → govern

governance → govern

The topic modeling system can therefore group these articles under a common topic related to **governing/public administration**.

Final Result

The morphology-based normalization mechanism identifies **govern** as the common root for all three terms.

- **govern** → **govern** → Base form → **govern**
- **government** → **govern + -ment** → Derivational → **govern**
- **governance** → **govern + -ance** → Derivational → **govern**

Thus, the final normalized representation for all three input words is:

govern

This normalization helps the news aggregation system group semantically related articles more effectively during **topic modeling and clustering**.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning. Apply the process to extract root forms and classify each transformation.
Determine the morphological breakdown and normalized output for each word.

A search optimization module processes words such as **“create”, “creates”, and “creating”**. These words have the same basic meaning but different grammatical forms. The system uses morphological normalization to identify their common root and remove inflectional suffixes.

The purpose is to ensure that a search for **create** can also retrieve documents containing **creates** or **creating**.

Step 1: Identify the Base Form

The common base word is:

create

The system checks each input word for an inflectional suffix.

Step 2: Apply Morphological Rules

Rule 1: Base Form

If the word has no inflectional suffix, retain it as it is.

create → create

Rule 2: Third-Person Singular

If a word ends in **-s**, remove the suffix.

creates → create + s → create

Here, **-s** represents a present-tense third-person singular form.

Rule 3: Progressive/Continuous Form

If a word ends in **-ing**, remove the progressive suffix and restore the base form.

creating → create + ing → create

The final **e** in *create* is dropped when forming *creating*, so the parser restores it during normalization.

Step 3: Morphological Breakdown

Word	**Root**	**Affix**	**Type**	**Stem**
create	create	—	Base	create
creates	create	-s	Inflectional	create
creating	create	-ing	Inflectional	create

Step 4: Classification

- **create** → Base form; no transformation.
- **creates** → **Inflectional**, because **-s** represents grammatical number/person and present tense.
- **creating** → **Inflectional**, because **-ing** represents progressive/continuous aspect.

Step 5: Normalized Output

create → create

creates → create

creating → create

Final Result

The search optimization module extracts the common root **create** from all three input words.

Final normalized representation: create

Thus, **create, creates, and creating** can be indexed under the same representation, improving search consistency while preserving their original meaning.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________